

SMITA RAJAN

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EDUCATION

University of California, Berkeley

Berkeley, CA

PhD Student, Department of Mathematics

August 2024–present

- Advisors: Hannah Larson and Bernd Sturmfels
- Visits at the Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany: June 2026, January 2026, June–July 2025, June–July 2024

Brown University

Providence, RI

Mathematics, Sc.B. with Honors; Mathematical Physics, Sc.B. with Honors

September 2020–May 2024

AWARDS AND HONORS

NSF Graduate Research Fellowship

2024–2029

University of California, Berkeley

Berkeley, CA

UC Berkeley Chancellor's Fellowship

2025–2026

University of California, Berkeley

Berkeley, CA

Albert A. Bennett Prize for Exceptional Accomplishment in Mathematics Concentration

2024

Brown University

Providence, RI

Mildred Widgoff Prize for Excellence in Thesis Preparation (Physics)

2024

Brown University

Providence, RI

PUBLICATIONS AND PREPRINTS

1. Universal formulas for de Jonquières Loci and Equivariant Contact Problems, arXiv:2609.30892.
2. Kinematic varieties for massless particles (with S. Sverrisdóttir and B. Sturmfels), *Le Matematiche*, 80 (2025), no. 1, 453–471, arXiv:2408.16711.
3. Five-dimensional spinor helicity for all masses and spins (with A. Pokraka, L. Ren, A. Volovich, and W. Zhao), *J. High Energy Phys.* (2025), no. 7, Paper No. 056, arXiv:2405.09533.

TALKS

2026

- Commutative Algebra and Algebraic Geometry Seminar, University of California, Berkeley
- Nonlinear Algebra Seminar, Max Planck Institute for Mathematics in the Sciences
- Representation theory learning seminar, Max Planck Institute for Mathematics in the Sciences
- Étale cohomology learning seminar, UC Berkeley
- Hodge theory learning seminar, UC Berkeley
- Algebraic Curves in January, Max Planck Institute for Mathematics in the Sciences

2025

- SIAM Conference on Applied Algebraic Geometry, University of Wisconsin–Madison
- Tropical geometry learning seminar, UC Berkeley

2024

- Geometric class field theory learning seminar, UC Berkeley
- Tensors learning seminar, UC Berkeley
- Commutative Algebra and Algebraic Geometry seminar, University of California, Berkeley
- Positive Geometry Seminar, Max Planck Institute for Mathematics in the Sciences

TEACHING

PAWS Problem Session Leader

September–November 2026

Arizona Winter School

Virtual

- Serving as a Problem Session Leader for the Preliminary Arizona Winter School (PAWS) session on “Algebraic Geometry via Computation,” led by Juliette Bruce. Duties include writing problem sets and meeting weekly with a small group of students.

Graduate Student Instructor

August–December 2024

University of California, Berkeley

Berkeley, CA

- Served as a Graduate Student Instructor for Math 1B. Duties included holding discussion sections, office hours, creating quizzes, and grading homework and exams.

Undergraduate Teaching Assistant

September 2021–May 2024

Brown University

Providence, RI

- Served as an Undergraduate Teaching Assistant for PHYS 0030, PHYS 0040, PHYS 0160, PHYS 1410, and PHYS 1420. Duties included facilitating problem-solving workshop sessions, holding weekly office hours, and grading.

MENTORING AND COMMUNITY INVOLVEMENT

IAS Women+ and Mathematics Ambassador

August 2026–May 2027

University of California, Berkeley

Berkeley, CA

- Helping strengthen the undergraduate math community at UC Berkeley by organizing workshops, panel sessions, speaker series, and community-building events.

Mentor for UC Berkeley Directed Reading Program

August 2025–June 2026

University of California, Berkeley

Berkeley, CA

- Spring 2026: Sheaf cohomology
- Fall 2025: Computational algebraic geometry
- Fall 2025: Introduction to scheme theory

Macaulay2 Workshop Leader

September 2025

Organizer

University of California, Berkeley

- Co-organized an introductory workshop for the UC Berkeley algebra community on how to use the computer algebra software Macaulay2.

Math Diversity and Inclusion Action Plan (DIAP) Committee

September 2023–May 2024

Member

Brown University

- Collaborated with a group of undergraduates, graduate students, and professors to address issues surrounding diversity, inclusion, and equity within the mathematics community at Brown.

Conference for Undergraduate Women in Physics

September 2022–January 2023

Local organizing committee member

Brown University

- Helped organize the 2023 APS Conference for Undergraduate Women in Physics at Brown University by planning workshops, panel sessions, career fairs, and logistics along with graduate students and professors.

TECHNICAL SKILLS

Programming Languages: Macaulay2, Mathematica, Python, SageMath, $\LaTeX$.